

RESEARCH EXPERIENCE

NSF EAR Postdoctoral Fellow Brown University, Department of Earth, Environmental and Planetary Sciences	starting Fall 2026
Senior Research Technician Duke University, Division of Earth and Climate Sciences	2026 – present
Graduate Student Researcher Duke University, Division of Earth and Climate Sciences	2020 – 2026
Research Assistant Stanford University, Department of Geophysics	2018 – 2020
Independent Research Assistant University of California, Santa Barbara, Department of Earth Sciences	2018 – 2019

EDUCATION

Duke University Ph.D., Earth and Climate Sciences Advisor: Avner Vengosh Dissertation: <i>The Aqueous Geochemistry of Lithium Deposits and Environmental Effects of Lithium Mining</i>	2020 – 2026
University of California, Santa Barbara B.S. Earth Science, Emphasis in Geohydrology	2015 – 2019

PUBLICATIONS

(*co-first author, †undergraduate mentee)

[Google Scholar Profile](#)

Submitted or In-Review

- [2] Wudke, H.; **Williams, G.D.Z.**; Vengosh, A., Baseline Water Quality of Shallow Groundwater used by Indigenous Communities near the Salar de Uyuni, Bolivia. *In-Revision at Applied Geochemistry*
- [1] Lu, P.L.; Harmon, R.S.; Leatherman, L.; **Williams, G.D.Z.**; Curry, A.C.; Kelly, L.; Dwyer, G.S.; Reynolds, V.S.; Dossou, A.; Richter, D.D., Behavior of Lithium during Soil Formation in Pegmatitic Systems: Lithium Signatures in Soil Minerals Derived from Li-Pegmatites. *In-Revision*

Published

- [19] Blake, M.; Butler, K.L.; Munk, L.A.; Boutt, D.F.; Morris, N.; Kennedy, J.; Saha, P.; Scotta, J.; Tompkins, H.; **Williams, G.D.Z.**; Vengosh, A.; Evaristo, J.; Ibarra, D.E., 2026. Reef-specific brine origin and lithium-enrichment mechanisms for the Devonian Leduc and Nisku Formations of the Alberta Basin, Canada. *Applied Geochemistry*. <https://doi.org/10.1016/j.apgeochem.2026.106973>
- [18] †Schaffer, A.; **Williams, G.D.Z.**; Dwyer, G.S.; Hyland, E.G.; Wong, J.M.; Osterberg, J.S.; Schultz, T.F.; Kipp, M.A., 2026. Environmental and vital effects on lithium isotopes and trace element ratios in biogenic carbonates: Insights from an estuarine salinity gradient. *Geochimica et Cosmochimica Acta*. <https://doi.org/10.1016/j.gca.2026.06.025>
- [17] Nativ, P.; **Williams, G.D.Z.**; Vengosh, A., 2026. A Simple, Field-Compatible Method for Accurate pH Measurement in Hypersaline Brines. *Applied Geochemistry* 206, 106890. <https://doi.org/10.1016/j.apgeochem.2026.106890>

- [16] **Williams, G.D.Z.**; Vengosh, A., 2026. Origins of the Mg/Li ratio in closed-basin brines of the Lithium Triangle: the relative importance of high- and low-temperature water-rock interactions. *Earth-Science Reviews* 278, 105516. <https://doi.org/10.1016/j.earscirev.2026.105516>
- [15] **Williams, G.D.Z.**; Barre, J.; Louvat, P.; Bérail, S.; Millot, R.; Vengosh, A., 2026. Geochemical controls on the formation of lithium brines in closed-basins of the Lithium Triangle. *Earth and Planetary Science Letters* 679, 119849. <https://doi.org/10.1016/j.epsl.2026.119849>
- [14] Nativ, P.; **Williams, G.D.Z.**; Vengosh, A., 2026. Discrepances Between pH and Corrosive Indexes of Hypersaline Effluents. *Environmental Science & Technology Letters* 13, 170–176. <https://doi.org/10.1021/acs.estlett.5c01196>
- [13] **Williams, G.D.Z.**; Petrović, M.; Hill, R.C.; †Hall, G.A.; Vengosh, A., 2025. The Water Quality Impacts of Legacy Hard-Rock Lithium Mining and Processing. *Environmental Science & Technology* 59, 26492–26505. <https://doi.org/10.1021/acs.est.5c13682>
- [12] *†Hall, G.A.; ***Williams, G.D.Z.**; Sirbescu, M.L.C.; Lu, P.L.; Dwyer, G.S.; Richter, D.D.; Vengosh, A., 2025. Strontium isotopes and Rb/Sr tracers in surface soils for locating subsurface lithium pegmatites. *Applied Geochemistry* 195, 106631. <https://doi.org/10.1016/j.apgeochem.2025.106631>
- [11] Hill, R.C.; Wang, Z.; Hu, J.; **Williams, G.D.Z.**; Vengosh, A., 2025. Radionuclides and the uranium isotope fingerprint of globally produced phosphate rocks, mineral fertilizers, and phosphogypsum waste and its potential effect on the environment. *Journal of Hazardous Materials* 499, 140033. <https://doi.org/10.1016/j.jhazmat.2025.140033>
- [10] Lopez, R.F.; Huayta, J.; **Williams, G.D.Z.**; Seay, S.A.; Lalwani, P.D.; Bacot, S.; Vengosh, A.; Meyer, J., 2025. Lithium Nickel Manganese Cobalt Oxide Particles Cause Developmental Neurotoxicity in *Caenorhabditis elegans*. *Environmental Science: Advances*. <https://doi.org/10.1039/D5VA00103J>
- [9] **Williams, G.D.Z.**; Nativ, P.; Vengosh, A., 2025. The role of boron in controlling the pH of lithium brines. *Science Advances* 11, eadw3268. <https://doi.org/10.1126/sciadv.adw3268>
- [8] **Williams, G.D.Z.**; Vengosh, A., 2025. Quality of Wastewater from Lithium-Brine Mining. *Environmental Science & Technology Letters* 12, 151–157. <https://doi.org/10.1021/acs.estlett.4c01124>
- [7] **Williams, G.D.Z.**; Saltman, S.; Wang, Z.; Warren, D.M.; Hill, R.C.; Vengosh, A., 2024. The potential water quality impacts of hard-rock lithium mining: Insights from a legacy pegmatite mine in North Carolina, USA. *Science of The Total Environment* 956, 177281. <https://doi.org/10.1016/j.scitotenv.2024.177281>
- [6] Hill, R.C.; Wang, Z.; **Williams, G.D.Z.**; Polyak, V.; Singh, A.; Kipp, M.A.; Asmerom, Y.; Vengosh, A., 2024. Reconstructing the depositional environment and diagenetic modification of global phosphate deposits through integration of uranium and strontium isotopes. *Chemical Geology* 662, 122214. <https://doi.org/10.1016/j.chemgeo.2024.122214>
- [5] Hill, R.C.; **Williams, G.D.Z.**; Wang, Z.; Hu, J.; El-Hasan, T.; Duckworth, O.W.; Schnug, E.; Bol, R.; Singh, A.; Vengosh, A., 2024. Tracing the Environmental Effects of Mineral Fertilizer Application with Trace Elements and Strontium Isotope Variations. *Environmental Science & Technology Letters* 11, 604–610 <https://doi.org/10.1021/acs.estlett.4c00170>
- [4] Hu, J.; Wang, Z.; **Williams, G.D.Z.**; Dwyer, G.S.; Gatiboni, L.; Duckworth, O.W.; Vengosh, A., 2024. Evidence for the accumulation of toxic metal(loid)s in agricultural soils impacted from long-term application of phosphate fertilizer. *Science of The Total Environment* 907, 167863. <https://doi.org/10.1016/j.scitotenv.2023.167863>
- [3] Wang, Z.; Hill, R.; **Williams, G.**; Dwyer, G.S.; Hu, J.; Schnug, E.; Bol, R.; Sun, Y.; Coleman, D.S.; Liu, X.-M.; Sandstrom, M.R.; Vengosh, A., 2023. Lead isotopes and rare earth elements geochemistry of global phosphate rocks: Insights into depositional conditions and environmental tracing. *Chemical Geology* 639, 121715. <https://doi.org/10.1016/j.chemgeo.2023.121715>

- [2] Vengosh, A.; Wang, Z.; **Williams, G.**; Hill, R.; M. Coyte, R.; Dwyer, G. S., 2022. The Strontium Isotope Fingerprint of Phosphate Rocks Mining. *Science of The Total Environment*, 850, 157971.
<https://doi.org/10.1016/j.scitotenv.2022.157971>
- [1] Wang, K.; Ellsworth, W. L.; Beroza, G. C.; **Williams, G.**; Zhang, M.; Schroeder, D.; Rubinstein, J., 2018, Seismology with Dark Data: Image-Based Processing of Analog Records Using Machine Learning for the Rangely Earthquake Control Experiment. *Seismological Research Letters*, 90 (2A), 553–562.
<https://doi.org/10.1785/0220180298>

INVITED PRESENTATIONS

University of Arizona, March 2026, *Geochemical Controls on the Origin, Exploration, and Environmental Impacts of Lithium Resources*

Indiana University Bloomington, Seminar on Critical Minerals and Climate Change Mitigation, December 2025, *Lithium Mining, Processing, and Environmental Impacts*

Goldschmidt Conference Session on Lithium Resources for the Energy Transition, July 2025, *The Many Roles of Boron in Controlling Lithium-Brine Geochemistry*. Abstract by Williams, G.D.Z. & Vengosh A.

CONFERENCE PRESENTATIONS & ABSTRACTS

Williams, GDZ.; Vengosh, A. (2025) [oral] The Role of Boron in Controlling the pH of Lithium Brines. The 3rd International Association of Geochemistry Conference.

Williams, GDZ.; Hall, GA.; Hill, RC.; Petrović, M.; Louvat, P.; Berail, S.; Barre, J.; Millot, R.; Warner, NR.; Baker, P.; Vengosh, A. (2025) [oral] The Origin of Lithium at the Salar de Uyuni: A multi-isotope and elemental approach. The 3rd International Association of Geochemistry Conference.

Williams, GDZ.; Vengosh, A. (2025) [oral] The role of boron in controlling the pH of evaporated lithium brines. AEESP Conference

Williams, GDZ.; Hill, RC.; Wang, Z.; Kipp, MA.; Vengosh, A. (2024) [poster] Searching for Alternative Critical Mineral Sources. Symposium on Critical Resources, Minerals, and Materials Joint Efforts

Williams, GDZ.; Hall, G.; Petrović, M.; Hill, RC.; Dwyer, G.; Vengosh, A. (2024) [oral] Water quality in a legacy lithium mining district of North Carolina. Goldschmidt Conference

Williams, GDZ. (2024) [oral] Geology and Production of Latin American Lithium Deposits. North Carolina Conference on Latin American Studies

Williams, G.; Wang, Z.; Hill, R.; Vengosh, A. (2023) [oral] Potential Water Quality Impacts of Hard-Rock Lithium Mining. AGU Fall Meeting

Williams, G.; Wang, Z.; Hill, R.; Vengosh, A. (2023) [poster] Water Quality and Legacy Lithium Mining in North Carolina: Insights on the Impacts of Future Lithium Mining. Goldschmidt Conference

Williams, G.; Vengosh, A.; Wang, Z.; Hill, R. (2022) [oral] Elemental Fluxes in Global Phosphate Ores: Evaluation of a Potential Resource for Critical Elements. GSA Connects

Williams, G.; Hill, R.; Wang, Z.; Vengosh, A. (2022) [poster] Multiple Isotopes as Potential Tracers for Contaminants Derived from Lithium Mine Wastes. GSA Connects

Williams, G.; Hill, R.; Wang, Z.; Whittaker, M.; Stringfellow, W.; Vengosh, A. (2022) [poster] Strontium Isotope Geochemistry as a Potential Tracer for Contaminants Derived from Lithium Mine Wastes. Goldschmidt Conference

ACADEMIC SERVICE & EXPERIENCE

Duke University, Earth & Climate Science Department Seminar Organizer & Discussion Leader, 2022-2024
Journal Guest Editor

- Applied Geochemistry (April 2026-present), *Geochemical Lenses into Environmental Legacies and Futures of Energy and Mineral Resources*. <https://www.sciencedirect.com/special-issue/332791/geochemical-lenses-into-environmental-legacies-and-futures-of-energy-and-mineral-resources>

Conference Session Co-Chair

- 2026 (upcoming), Goldschmidt, Montréal, Canada; C Chen, T Martins, F Putzolu, **GDZ Williams** – *Lithium resources across Earth systems*
- 2026 (upcoming), Goldschmidt, Montréal, Canada; RM Coyte, Z Wang, A Vengosh, **GDZ Williams** – *Geochemical lenses into the environmental legacies and futures of energy and mineral resources*
- 2025, The 3rd International Association of Geochemistry Conference, Cagliari, Italy; E Sacchi, **GDZ Williams** – *Isotopic Tools and Applications for Sustainable Water Management, Mining-Related Settings, Environmental Sciences, and the Energy Transition*

Reviewer: *Geochimica et Cosmochimica Acta*, Environmental Science & Technology, Applied Geochemistry

Society Affiliations: Geochemical Society, International Association of Geochemistry, American Geophysical Union, Geological Society of America

GRANTS & AWARDS

NSF EAR Postdoctoral Fellowship, starting Fall 2026 at Brown University

Global Student Research Fund, Duke Office of Global Affairs (**\$2,145**), 2025

Fall Tuition Scholarship, Duke Graduate School (**\$5,215**), 2025

Summer Research Fellowship, Duke Graduate School (**\$14,850**), 2025

Chateaubriand Fellowship, French Embassy to the USA (**€7,967**), to conduct lithium isotope work in Pau, France at UPPA, 2024

Dissertation Research Travel Award: International, Duke Graduate School (**\$5,000**), 2023-2024

Dissertation Research Travel Award: Domestic, Duke Graduate School (**\$4,000**), 2023-2024

Potential Impacts of Lithium Mining on Water Quality in North Carolina, NC Water Resources Research Institute (**\$120,000 direct**), PI Avner Vengosh (written with Gordon Williams), 2023-2025

The Potential Environmental Effects of Lithium Mining and Extraction, Albemarle Corporation (**\$396,230 direct**), G. Williams, A. Vengosh, D. Shindell, 2023-2024

Graduate Student Research Grant, Geological Society of America (**\$1,749**), 2022

Student Travel Grant, Geological Society of America, 2022

TEACHING & MENTORING

Teaching (Instructor of Record)

Critical Minerals Nexus: Geosciences, Engineering and Policy Spring 2026, Bass Connections project class co-led with Dr. Erika Weinthal

Graduate Teaching Assistant

EOS 101 Dynamic Earth with Dr. Emily Klein, Fall 2020

EOS 101 Dynamic Earth with Dr. Alex Glass, Spring 2021

EOS 220 Water Sciences with Dr. Avner Vengosh, Fall 2021

EOS 524 Water Quality and Health with Dr. Avner Vengosh, Spring 2022

ECS 401 Geology of North Carolina (with multi-day field trips) with Dr. Alex Glass, Fall 2022

ECS 201L Earth Materials with Dr. Adam Curry, Spring 2023

ECS 410 Senior Capstone Experience, The Geology of Ireland (with 10-day field trip) with Dr. Gary Dwyer, Spring 2023 & Spring 2025

Guest Lectures & Field Trip Demonstrations/Presentations

ECS 524, Water Quality and Health
ECS 201L, Earth Materials
ECS 210S, Exploring Earth Science: Field and Laboratory Investigations
ECS 226S, Field Methods in Earth and Environmental Sciences

Mentored Master's Student Research

1. Yitong (Eric) Su, 2025-present, Developing isotopic tracers for determining the provenance of critical raw materials
2. Sam Saltman, 2022-2023, Investigating the potential environmental impact of lithium mining in North Carolina
3. Ryan Parks, 2022-2023, Investigating the current and historical groundwater quality at the Maplewood Cemetery and Pauli Murray Center

Mentored Undergraduate Research

1. Aliza Savin, 2026-present, project: The water quality impacts of commercial cranberry bogs
2. Alexandra Schaffer, 2025-present. project: Understanding the lithium isotope and trace metal geochemistry of oyster shells
3. Grace Hall, 2023-2025. thesis: "The Potential Water Quality Impacts of Mining Critical Raw Materials: Comparative Analysis of Simulated Leachates of Sulphide and Laterite Ores"
4. Tiana Dinham, 2023, project: Investigating the water quality impacts of hard-rock lithium mining in North Carolina
5. Carsten Pran, B.S. (2022), thesis: "An Interdisciplinary Approach to Understanding Duke University's Relationship with the Durham County Water System"
6. Bass Connections Project (2021) with an undergraduate team of 5 students: "Inspecting the Taps: Radium, Groundwater, and Public Health in the Heart of Texas"

PROFESSIONAL REFERENCES

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